Name ______________________ Student ID ______________________

UCCC 2063 Algorithm Analysis

Time allowed: 1 hour 20 minutes

Full marks = 70

Put down your name and ID at the top of ALL pages.
Answer ALL the questions.

Question One: [Solving Recurrences]
Express your answer in Big-Oh notation. Your answer must be tight.

a) Solve T(n) = 4 T(n / 6) + n, by master method. (10 marks)
b) Solve T(n) = 4 T(n / 6) + n, by expansion (or recursive method). (10 marks)
c) Solve T(n) = 4 T(n / 6) + n, by MI (or substitution). (10 marks)
d) Solve T(n) = 8 T(n / 6) + n, by master method. (10 marks)
e) Solve T(n) = 8 T(n / 6) + n, by expansion (or recursive method). (10 marks)
f) Solve T(n) = 8 T(n / 6) + n, by MI (or substitution). (10 marks)

Question Two: [Usability of Master Method]

a) Given f(n) = n / log n, a ≥ 1, b > 1, and n is any very large positive number, prove

$$\forall k \geq 0 \bullet f(n) \neq \Theta(n^{\log_b a} \log^k n).$$

*** No need to go into the definition of Θ. (5 marks)

b) Briefly explain why it is NOT feasible to use master method for solving the recurrence T(n) = 2T(n / 2) + n / log n. (5 marks)

Answer: (may use two-column style)

Name

Student ID